CaseFlow — Backend Schema Specification

Version 2.5 · Database Architecture, DDL & Security Policies

Confidential — For Database Administrators & Backend Engineers

1. Relational Database Overview

CaseFlow's persistence engine is built on **PostgreSQL 15** via Supabase. The database consists of **38 tables** organized into 6 core logical domain schemas: 1. **Core Practice Management** (`profiles` , `clients` , `cases` , `hearings` , `documents` , `payments` , `user_roles`) 2. **Case Strategy Board** (`strategy_tasks` , `task_comments` , `task_attachments` , `task_activity_log`) 3. **Local Commission Module** (`commissions` , `commission_visits` , `commission_observations` , `commission_evidence` , `commission_expenses` , `commission_reports` , `commission_witnesses` , `commission_sketches` , `commission_notifications`) 4. **Bar Association Platform** (`bar_associations` , `bar_members` , `bar_admins` , `bar_notices` , `bar_notice_reads` , `bar_fee_payments` , `bar_elections` , `bar_election_posts` , `bar_nominations` , `bar_votes` , `bar_voter_participation` , `bar_events`) 5. **Legal & Citizen Services** (`bare_act_sections` , `caveats` , `certified_copies` , `ai_chat_sessions`) 6. **Telemetry & System** (`audit_logs` , `carousel_slides` , `fcm_tokens` , `device_tokens`)

2. Entity Relationship Diagram (ERD)

```
erDiagram
    PROFILES ||--o{ CASES : "creates / assigned"
    CLIENTS ||--o{ CASES : "owns"
    CASES ||--o{ HEARINGS : "has"
    CASES ||--o{ DOCUMENTS : "contains"
    CASES ||--o{ PAYMENTS : "records"
    CASES ||--o{ STRATEGY_TASKS : "manages"
    CASES ||--o{ COMMISSIONS : "orders"
    STRATEGY_TASKS ||--o{ TASK_COMMENTS : "has"
    STRATEGY_TASKS ||--o{ TASK_ATTACHMENTS : "includes"
    COMMISSIONS ||--o{ COMMISSION_VISITS : "conducts"
    COMMISSIONS ||--o{ COMMISSION_EVIDENCE : "gathers"
    BAR_ASSOCIATIONS ||--o{ BAR_MEMBERS : "enrolls"
    BAR_ASSOCIATIONS ||--o{ BAR_ELECTIONS : "holds"
    BAR_ELECTIONS ||--o{ BAR_NOMINATIONS : "receives"
    BAR_ELECTIONS ||--o{ BAR_VOTES : "collects"
```

3. Data Dictionary & Table Definitions

3.1 Core Practice Schema

`profiles` (User & Advocate Directory) Stores advocate profiles, firm links, and subscription parameters.

```
CREATE TABLE public.profiles (
    id                          UUID PRIMARY KEY REFERENCES auth.users(id) ON DELETE CASCADE,
    full_name                   TEXT NOT NULL DEFAULT '',
    email                       TEXT,
    user_role                   TEXT NOT NULL DEFAULT 'advocate',
    subscription_tier           TEXT DEFAULT 'free',
    firm_id                     UUID,
    bar_association_id          UUID REFERENCES public.bar_associations(id) ON DELETE SET NULL,
    registration_number         TEXT,
    practice_state              TEXT,
    practice_district           TEXT,
    office_address              TEXT,
    mobile_number               TEXT,
    is_admin                    BOOLEAN DEFAULT false,
    is_public_directory_enabled BOOLEAN DEFAULT true,
    last_seen_at                TIMESTAMPTZ,
    created_at                  TIMESTAMPTZ NOT NULL DEFAULT now(),
    updated_at                  TIMESTAMPTZ NOT NULL DEFAULT now()
);
```

`cases` (Primary Case Ledger)

```
CREATE TABLE public.cases (
    id               UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    client_id        UUID NOT NULL REFERENCES public.clients(id) ON DELETE CASCADE,
    case_title       TEXT NOT NULL,
    case_number      TEXT,
    cnr_number       TEXT UNIQUE,
    case_type        TEXT,
    court_name       TEXT,
    stage            TEXT NOT NULL DEFAULT 'filing',
    status           TEXT NOT NULL DEFAULT 'active',
    filing_date      DATE,
    next_hearing_date DATE,
    assigned_to      UUID REFERENCES auth.users(id),
    firm_id          UUID,
    created_by       UUID NOT NULL REFERENCES auth.users(id) ON DELETE CASCADE,
    deleted_at       TIMESTAMPTZ,
    created_at       TIMESTAMPTZ NOT NULL DEFAULT now()
);
```

`hearings` (Court Diary & Adjournments)

```
CREATE TABLE public.hearings (
    id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    case_id       UUID NOT NULL REFERENCES public.cases(id) ON DELETE CASCADE,
    hearing_date  DATE NOT NULL,
    stage         TEXT,
```

```
    notes          TEXT,
    next_action    TEXT,
    adjourn_reason TEXT,
    is_completed   BOOLEAN DEFAULT false,
    created_by     UUID NOT NULL REFERENCES auth.users(id) ON DELETE CASCADE,
    created_at     TIMESTAMPTZ NOT NULL DEFAULT now()
);
```

3.2 Case Strategy Board Schema

`strategy_tasks` (Kanban Task Cards)

```
CREATE TABLE strategy_tasks (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    case_id           UUID NOT NULL REFERENCES cases(id) ON DELETE CASCADE,
    title             TEXT NOT NULL,
    description       TEXT,
    stage             TEXT NOT NULL DEFAULT 'todo'
                      CHECK (stage IN ('todo', 'in_progress', 'review', 'filed', 'done')),
    position          INTEGER NOT NULL DEFAULT 0,
    assigned_to       UUID REFERENCES profiles(id) ON DELETE SET NULL,
    created_by        UUID NOT NULL REFERENCES profiles(id),
    priority          TEXT NOT NULL DEFAULT 'medium'
                      CHECK (priority IN ('low', 'medium', 'high', 'urgent')),
    due_date          DATE,
    linked_hearing_id UUID REFERENCES hearings(id) ON DELETE SET NULL,
    firm_id           UUID,
    is_auto_generated BOOLEAN DEFAULT false,
    completed_at      TIMESTAMPTZ,
    deleted_at        TIMESTAMPTZ,
    created_at        TIMESTAMPTZ DEFAULT now(),
    updated_at        TIMESTAMPTZ DEFAULT now()
);
```

---

4. Row Level Security (RLS) Policy Engine

CaseFlow mandates strict tenant isolation via Row Level Security.

4.1 Multi-Tenant Firm Isolation Strategy

```
-- RLS Policy: Selective Access to Strategy Tasks
CREATE POLICY "strategy_tasks_select" ON strategy_tasks
    FOR SELECT USING (
        created_by = auth.uid()
        OR assigned_to = auth.uid()
        OR firm_id = auth.uid()
        OR firm_id = (SELECT firm_id FROM profiles WHERE id = auth.uid())
    );

CREATE POLICY "strategy_tasks_insert" ON strategy_tasks
    FOR INSERT WITH CHECK (created_by = auth.uid());
```

---

5. Storage Bucket Definitions

Bucket ID	Access Control	Allowed MIME Types	File Size Limit
`case-documents`	Private (RLS)	`application/pdf`, `image/`	50 MB
`chat-attachments`	Public Read	`application/pdf`, `image/`, `audio/*`	25 MB
`commission-evidence`	Public Read	`image/jpeg`, `image/png`	15 MB
`commission-sketches`	Public Read	`image/png`	10 MB
`profile-photos`	Public Read	`image/jpeg`, `image/png`	5 MB
`bare_acts`	Public Read	`application/pdf`	100 MB

End of Backend Schema Specification · CaseFlow Engineering Group